

Homework 3

TEMBA Statistics 2026

This assignment covers portfolios, the central limit theorem, sampling, confidence intervals, and tests for means and proportions. It is graded **credit/no credit**.

Set up the workspace. Download and unzip `IntroStatGPT_homework_03_2026-09-28.zip` from the course IntroStatGPT page. Open the unzipped `TEMBA-2026-homework-03-codex` folder in Codex, following the [course setup instructions](#). Ask: “Set up this IntroStatGPT workspace and verify that it is ready. Do not begin an analysis yet.”

The workspace includes this assignment and four datasets:

Dataset	Rows
<code>portfolio_monthly_returns</code>	192
<code>bloom_attrition</code>	1,612
<code>bloom_satisfaction</code>	1,346
<code>bloom_performance</code>	1,507

Problems 2–4 supply their numerical inputs in the questions.

Assemble and submit your work. Save the requested tables and plots as you work. Ask IntroStatGPT to arrange them in a Word report in problem order, review its proposed outline, and approve the build. Add your calculations and written answers under the corresponding problem and part numbers, then export to PDF. You may submit one or more PDFs; legible scans or photographs of handwritten work are acceptable once converted to PDF. Check that every part is included and all tables, plots, calculations, and answers are readable.

The course IntroStatGPT page also has an optional Homework 3 sample chat and report skeleton. Use them to check your outputs or as a backup if the tool is unavailable. Write your own explanations and recommendation.

Use IntroStatGPT for the requested tables, plots, and software checks. For hand calculations, show your formula, substituted values, and units before requesting a check. For normal probabilities at one, two, or three SDs from the mean, use the 68/95/99.7% rule. For other standardized values, IntroStatGPT may evaluate normal probabilities once you supply the standardized value and the tail or tails. Write your own interpretations.

Use the methods from the notes for hand calculations of confidence intervals and tests, with a 5% significance level. Software results may differ slightly because of small technical adjustments; compare the conclusions as well as the numbers. Small differences from rounding are acceptable. Problems 2–4 use hypothetical numbers.

Problem 1. Portfolio returns

The file `portfolio_monthly_returns.csv` contains 192 monthly returns for **SPY**, a fund tracking the S&P 500, and **GLD**, a fund holding gold, from January 2010 through December 2025. Each row pairs the returns for one month. Returns use prices adjusted for splits and distributions and are in decimal units: 0.02 means a 2% gain and -0.03 means a 3% loss.

Invest a fraction w in SPY and $1 - w$ in GLD, rebalancing to these weights at the start of each month. Ignoring trading costs and taxes, the monthly portfolio return is

$$R_P = wR_S + (1 - w)R_G,$$

where R_S and R_G are that month's SPY and GLD returns. Use the historical means, sample SDs, and covariance to estimate the portfolio's expected return and SD.

1.1. Return summaries

Ask IntroStatGPT for the observation count, mean, and sample SD of each return, and the correlation and covariance between them. For later hand calculations, retain six decimal places for means, SDs, and correlation, and eight for covariance, using decimal returns. Report final expected returns as percentages and SDs in percentage points, both to two decimal places; report correlations to three decimal places.

Also request a scatterplot of GLD return against SPY return for the full period.

- What does each point represent? Describe the relationship and one feature of the plot that correlation alone misses.
- Verify that covariance equals correlation times the two SDs. Explain the units of covariance and correlation.
- Does correlation near zero establish independence? Explain.

1.2. A 60/40 portfolio

Consider an allocation of 60% SPY and 40% GLD.

- Calculate the estimated expected monthly return and SD. Show the covariance term in the variance calculation.
- Compare your SD with $0.60s_S + 0.40s_G$. What relationship between the returns would make that weighted average the correct portfolio SD?
- Ask IntroStatGPT to calculate each month's 60/40 portfolio return, then take the mean and sample SD of those returns. Compare with your hand calculation. The results should agree apart from rounding.

1.3. Portfolio allocations

Assume a fixed risk-free return of **0.25% per month**, or $r_f = 0.0025$. The estimated monthly **Sharpe ratio** is

$$\frac{\text{estimated expected monthly return} - r_f}{\text{monthly return SD}}.$$

It measures expected return above the risk-free return per unit of SD. Use portfolios containing only SPY and GLD, with $0 \leq w \leq 1$.

Ask IntroStatGPT for a three-panel plot showing expected return, SD, and Sharpe ratio against the percentage invested in SPY. Have it find the allocation with the highest Sharpe ratio over the full range from 0% to 100% SPY and mark that allocation on all three panels. Report its weights, expected return, SD, and Sharpe ratio. Also report those quantities for the minimum-SD allocation. Keep all calculations monthly. Report allocation percentages to one decimal place and Sharpe ratios to four decimal places. Round after finding the optimum.

- Explain the shapes of the expected-return and SD curves. Why can a mix of the two assets have a lower SD than either asset alone?
- Compare the maximum-Sharpe, minimum-SD, and 60/40 portfolios. What tradeoffs do they make between expected return and SD?
- Repeat the return summaries and find the maximum-Sharpe allocation separately for 2010–2017 and 2018–2025, using the same risk-free return. How do the means, SDs, correlation, and optimal weights change between periods? Why might the allocation that maximizes the historical Sharpe ratio fail to maximize it for future returns?
- Consider two portfolios holding different assets, with the same expected monthly return and SD. Could they have different probabilities of losing more than 10% in a month? Would their Sharpe ratios distinguish that risk? Explain.

Problem 2. Lunch orders

A festival food stand will serve 100 customer groups. The number of lunches purchased per group has mean 3 and SD 2. Treat the groups' purchases as iid. Individual purchases are right-skewed; assume the CLT gives an adequate normal approximation for total demand.

- Find the mean and SD of total lunch demand. Why can total demand be approximately normal when individual purchases are right-skewed?
- If the stand prepares 320 lunches, use the 68/95/99.7% rule to approximate the probability that demand exceeds its stock.
- Use the mean-plus-two-SDs rule to choose a stock that meets demand about 97.5% of the time.
- At a larger festival, the stand will serve 400 groups with the same purchase distribution. The manager proposes ordering four times the quantity in part c. Apply the same stocking rule to 400 groups and compare. Explain why expected demand and the extra stock above it grow at different rates.

Problem 3. Inspection workload

A rental-equipment hub has 900 staff-minutes available tomorrow. It must process 100 routine returns and can book 32, 34, or 36 additional inspections. Processing times have these means and SDs:

Job	Mean time	SD of time
Routine return	4 minutes	2 minutes
Inspection	12 minutes	8 minutes

Times are independent across jobs and identically distributed within each job type. Assume each type's total time is approximately normal. Plan using total staff-minutes; ignore the timing of arrivals.

- For 34 inspections, calculate the mean and SD of the total workload. Show how routine returns and inspections contribute to each calculation.
- Approximate the probability of exceeding 900 minutes for each booking size. Show the mean, SD, and standardized capacity in a table. What is the largest booking with no more than a 5% chance of exceeding capacity?
- Why is an expected workload below 900 minutes insufficient to meet that requirement? Use one of the booking sizes to illustrate.

Problem 4. Consumer survey

This simplified survey borrows topics from the University of Michigan's [Surveys of Consumers](#). The population consists of 250 million adults on a fixed date. Researchers select a simple random sample of 400 adults. Everyone selected responds accurately. Treat the population's answers as fixed throughout this problem.

The same 400 people answer both questions:

Survey item	Sample result
Expected percentage change in prices over the next 12 months	Mean 3.4%; sample SD 4.0 percentage points.
Whether their family's finances are worse than a year ago	208 say worse; 192 do not.

Use the standard error formulas from the notes and assume normal approximations are adequate.

4.1. Inflation expectations

- Identify the observational unit, population, sample, and population mean being estimated.
- Calculate the SE and our approximate 95% confidence interval for the population mean expectation. Explain why neither statement correctly describes the interval: "It tells us where inflation will be next year" and "It contains about 95% of adults' expectations."
- Test whether the population mean expectation is 4%, against a two-sided alternative. Use your SE from part b to calculate the t-statistic and the 68/95/99.7% rule to approximate the two-sided p-value. Ask IntroStatGPT to check the test and report a 95% confidence interval using the supplied sample size, mean, and SD. Do the conclusions agree?
- Imagine independently drawing 1,000 samples of 400 from this same population, keeping everyone's answers fixed. Each draw starts from the full population. What stays fixed and what varies? About how many of our approximate 95% confidence intervals would contain the population mean? Why would surveys taken in different months be a different exercise?

4.2. Family finances

Let p be the population fraction who would answer “worse.”

- Estimate p , calculate its SE, and give our approximate 95% confidence interval. What does the interval say about whether a majority of the population would answer “worse”?
- Test $H_0 : p = 0.50$ against $H_A : p \neq 0.50$. Use your SE from part a to calculate the t-statistic and approximate normal p-value. Which sample results count as at least as extreme as the observed result for this test?
- If the sample size increased to 1,600 and the sample proportion stayed near 52%, what would happen to the SE and margin of error?

Problem 5. Hybrid work

Bloom, Han, and Liang studied 1,612 university-graduate employees in Trip.com’s Airfare and IT divisions during a six-month experiment in 2021–2022. The researchers randomly assigned employees by birthday: odd-numbered birthdays received the option to work at home Wednesday and Friday; even-numbered birthdays worked in the office five days a week. Employees offered hybrid work chose how much to use it.

The supplied data contain three outcomes by assigned group:

Dataset	Outcome
bloom_attrition	All assigned employees; quit is 1 if the employee quit during the experiment and 0 otherwise.
bloom_satisfaction	Survey responses measuring work satisfaction on a 0–10 scale.
bloom_performance	Available performance reviews, coded D = 1, C = 2, B = 3, B+ = 4, A = 5.

Some employees lack survey responses or performance reviews. In all comparisons, subtract the office result from the hybrid result and use the number with an observed outcome for each group’s calculations. Use the notes’ standard error formulas for comparisons of independent groups as approximations here.

5.1. Study design and summaries

What does comparing the assigned groups tell us about offering hybrid work? Why would comparing employees who actually worked at home with those who did not answer a different question? How does random assignment here differ from random sampling in Problem 4?

Ask IntroStatGPT for one summary table by assigned group and outcome. Include:

- Report the number assigned, the number with an observed outcome, and the number missing that outcome (assigned minus observed).
- Report the quitting count and proportion, and the mean and sample SD for satisfaction and performance.

Use four decimals for means and SDs and six for proportions. Missing outcomes include employees absent from a file and blank values within it.

5.2. Quitting

- Calculate the hybrid group's quit proportion minus the office group's quit proportion, its SE, and our approximate 95% confidence interval. Report the difference and interval in percentage points. Then express how much lower the hybrid group's quit proportion is as a percentage of the office group's quit proportion. Why do these two ways of describing the reduction give different numbers?
- State the null hypothesis of equal quit probabilities and a two-sided alternative. Use your SE from part a to calculate the t-statistic and approximate normal p-value. What do you conclude at the 5% significance level?
- Explain why this claim is wrong: "The p-value is the probability that offering hybrid work has no effect on quitting." Give a correct interpretation of the p-value in this study.

5.3. Performance and satisfaction

- Calculate the hybrid group's mean performance minus the office group's mean performance, its SE, and our approximate 95% confidence interval. Interpret the interval in points on the supplied 1–5 performance scale.
- Ask IntroStatGPT to compare the groups' mean performance and satisfaction, reporting each estimated difference, a 95% confidence interval, and a two-sided p-value for testing a mean difference of zero. Compare the performance interval with yours and interpret the satisfaction estimate and interval.
- Which of these differences in mean performance are compatible with the interval: zero, a hybrid mean 0.50 points below the office mean, and a hybrid mean 0.10 points below the office mean? Does including zero establish that the means are equal?
- Why doesn't the smaller p-value for satisfaction establish that its benefit matters more to the business than the reduction in quitting?

5.4. Recommendation

Write a short recommendation about offering hybrid work, using the estimates and uncertainty for all three outcomes. Explain how missing outcomes could limit the comparisons and identify one limitation on applying the results to another employer.

Sources

- SPY and GLD historical adjusted prices, [Yahoo Finance](#), downloaded on September 27, 2026. Returns use the last trading day of each month. The risk-free return in Problem 1 is hypothetical.
- University of Michigan, Survey Research Center, [Surveys of Consumers: Survey Description](#). Problem 4 borrows survey topics; its population, sampling design, and numerical results are invented for the exercise.
- Bloom, N., Han, R., and Liang, J. (2024), "[Hybrid working from home improves retention without damaging performance](#)", *Nature* 630, 920–925; [replication data](#).